

Autodesk® Scaleform®

Scaleform 4.3 AS2 Extensions Reference

Scaleform 4.3

ActionScript 2.0

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: 4.03
: 2012 9 17

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Autodesk® Scaleform® 4.3

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Autodesk Scaleform :

ActionScript Extensions Reference (ActionScript
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1 . @ *

Autodesk Scaleform SDK™
PC
Scaleform Adobe Flash Studio

Flash®

Flash Web

Scaleform ActionScript IME

_global.gfxExtensions true
Scaleform Player

_global.gfxExtensions = true;

FLA
Scaleform Player

Flash

Flash® Player

Scaleform

Scaleform
SDK
Player

Scaleform Player
Scaleform

Scaleform API

API

Scaleform

2 Mouse a > k - Z a k x ¥ i ™ ¥

Mouse
 (16)
 Mouse
 Mouse
 Scaleform
 Scaleform
 Flash

2.1 3 / -] 4 o • - g • 4 z

Wii
 Scaleform 4.0 4
 GfX::Movie::SetMouseCursorCount(unsigned n)
 GfX::MouseEvent GfX::Movie::HandleEvent
 Scaleform GfX::MouseEvent
 GfX::MouseEvent

2.2 Button V < ¥ z - Z a k x ¥ i ™ ¥

onRollOver onRollOut onDragOver onDragOut onPress onRelease onReleaseOutside
 Button Scaleform 1 2

onPress 1
 1 :

```

mc.onPress = function(mouseIdx:Number)
{
    if (mouseIndex == 0)
        ...
    else if (mouseIndex == 1)
        ...
    ...
}
  
```

onPress onRelease 2
 NUMERIC -1 0

```

mc.onPress = function(mouseIdx:Number, keyboardOrMouse:Number)
  
```

```

{
    if (keyboardOrMouse == 0)
    else
    ...
}

2
onRollOver/Out
onDragOver/Out
rollover/dragover
onDragOver/onDragOut
onRollOver/onRollOut
onRollOver/onDragOver
onRollOut/onDragOut

2
onRollOver/onRollOut
onDragOver/onDragOut
(
    onRollOver/onDragOver
    2
    onRollOver/onRollOut
    0
    2
    1
    2
    onRollOut/onDragOut
    1
    onRollOut/onDragOut
    0
    2
    2
    onRollOver/onRollOut
    2
    onDragOver/onDragOut

mc.onRollOver = function(mouseIdx:Number, nestingIdx:Number)
{
    // nestingIdx == 0
    if (nestingIdx == 0)
        doOverAnimation();
    ...
}
mc.onRollOut = function(mouseIdx:Number, nestingIdx:Number)
{
    // nestingIdx == 0
    if (nestingIdx == 0)
        doOutAnimation();
    ...
}

on(rollOver) on(rollOut)

```

onPress onRelease

Button Events

- x onPressAux
- x onReleaseAux
- x onReleaseOutsideAux
- x onDragOver
- x onDragOut

index != 0

0

```
mc.onPressAux = function(mouseIdx:Number, keyboardOrMouse:Number, buttonIdx:Number)
{
    if (buttonIdx == 1) // Right mouse button
        ...
    ...
}
```

2.3 Mouse a > k V < ¥ z

ActionScript

Flash

Mouse.addListener
:

```
f onMouseMove = function() {}
f onMouseDown = function() {}
f onMouseUp = function() {}
f onMouseWheel = function(delta : Number, targetPath : String) {}
```

Flash

onMouseDown

onMouseUp

(
 _global.gfxExtensions true) Scaleform

:


```
f onMouseDown = function(button : Number, [targetPath : String],
    [mouseIdx : Number], [x : Number], [y : Number] ,
    [dbClick : Boolean]) { }
f onMouseUp = function(button : Number, [targetPath : String],
    [mouseIdx : Number], [x : Number], [y : Number]) { }
```

1 Scaleform
2 3 16 1
Flash
:

```
var mouseListener:Object = new Object;

mouseListener.onMouseDown = function(button, target)
{
    trace("mouseDown      - button " + button +
        " target = " + target + "");
}
mouseListener.onMouseUp = function(button, target)
{
    trace("mouseUp      - button " + button +
        " target = " + target + "");
}

Mouse.addListener(mouseListener);
```

mouseDown/Up button target 2
button Mouse["LEFT"] Mouse["RIGHT"]
Mouse["MIDDLE"] button
(Mouse["LEFT"] ActionScript 2.0 (target)
Mouse.LEFT) 2
target
'undefined'
Flash 'undefined'
'undefined' Flash Player

```
f onMouseMove = function([mouseIdx : Number],
    [x : Number], [y : Number]) { }
f onMouseDown = function(button : Number, [targetPath : String],
```

```

        [mouseIdx : Number], [x : Number], [y : Number] ,
        [dbClick : Boolean]) { }
    f onMouseUp = function(button : Number, [targetPath : String],
        [mouseIdx : Number], [x : Number], [y : Number]) { }
    f onMouseWheel = function(delta : Number, targetPath : String,
        [mouseIdx : Number], [x : Number], [y : Number]) { }

    mouseIdx

```

```

x y      (_root      )
onMouseDown      dbClick
                'true'

```

2.4 • X k] 4 o • - × Æ

Flash :

```

f      -
f      - useHandCursor      true
f l      -

```

Flash Scaleform

C++ API ActionScript Mouse
C++ GFX::StateBag::SetUserEventHandler

FxPlayer.cpp

ActionScript

SDK

Mouse.fla/swf

Mouse

'Cursor_M'

"hand"

"arrow" "ibeam"

1

_x _y

Mouse

Scaleform

Mouse.setCursorType
cursorType mouseIndex
Mouse
cursorType

```
(Mouse["ARROW"] Mouse["HAND"] Mouse["IBeam"])
```

Mouse

```
_global.gfxExtensions = 1;
//
Mouse.hide();

var mouseListener:Object = new Object;
mouseListener.onMouseMove = function(mouseIdx, x,y)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    if (x == undefined)
    {
        x = _xmouse;
        y = _ymouse;
    }
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));

    cmc._x = x;
    cmc._y = y;
}
Mouse.addListener(mouseListener);

Mouse["setCursorType"] = function(cursorType, mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    switch(cursorType)
    {
        case Mouse["HAND"]:
            cmc.Cursor.gotoAndPlay("hand");
            break;
        case Mouse["ARROW"]:
            cmc.Cursor.gotoAndPlay("arrow");
            break;
        case Mouse["IBeam"]:
            cmc.Cursor.gotoAndPlay("ibeam");
            break;
        default:
            return;
    }
}
```

```

// Mouse.show   F Mouse.hide
//
ASSetPropFlags(Mouse,      "show,hide", 0, 7);

Mouse.show = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = true;
}
Mouse.hide = function(mouseIdx)
{
    if (mouseIdx == undefined)
        mouseIdx = 0;
    var cmc = eval("_root.Cursor_M"+(mouseIdx+1));
    cmc._visible = false;
}

```

2.5 Mouse a > k - ActionScript Z a k x ¥ i ™ ¥

Mouse ActionScript
 ActionScript 1.0 Mouse.setCursorType(...)
 ActionScript 2.0
 ActionScript 2.0
 Mouse["setCursorType"](..)

getButtonsState() ¯ Ó ¨ o u {

public function getButtonsState(mouseIndex : Number) : Number

Scaleform : 2.2

Æ Û Ó î ±
 mouseIndex : Number -

getTopMostEntity () ¯ Ó ¨ o u {

```

public function getTopMostEntity([mouseIndex :      Number]) :      Object
public function getTopMostEntity(testAll :      Boolean,
                                   [mouseIndex :      Number]) :      Object
public function getTopMostEntity(x :      Number, y :      Number) :      Object
public function getTopMostEntity(x :      Number, y :      Number, testAll:      Boolean) :      Object

```

Scaleform : 1.2.34 2.2

(onPress onMouseDown onRollOver)

```

MovieClip.hitTest      hitTest x y
getTopMostEntity      x y
Mouse.getTopMostEntity(x, y).hitTest(x, y,
true) == true

```

```

    mouseIndex :      Number      -
    testAll :      Boolean      -      (false)
    (true)
getTopMostEntity      'true'
    x :      Number, y : Number      -

```

```

[:

var listenerObj = new Object;

listenerObj.onMouseMove = function()
{
    var target = Mouse["getTopMostEntity"]();
    trace("Mouse moved, target = "+target);
}

Mouse.addListener(listenerObj);

var target = Mouse["getTopMostEntity"](480, 10);
trace("The character at the (480, 10) is " + target);

```

```

getPosition      " o u {

public function getPosition(mouseIndex :      Number) :      flash.geom.Point

Scaleform      : 2.2

```

flash.geom.Point

ÃÓ±

mouseIndex : Number –

setCursorType() – Ó “ o u {

public function setCursorType(cursorType : Number, [mouseIndex : Number]) : void

Scaleform : 1.2.32

cursorType

ÃÓ±

cursorType : Number – Mouse.ARROW Mouse.HAND Mouse.IBEAM

mouseIndex : Number –

3 Button a > k Z a k x ¥ i ™ ¥

hitTestDisable % º Ÿ f x U

hitTestDisable:Boolean [read- write]

Scaleform : 2.1.51

true MovieClip.hitTest

false

:
TextField.hitTestDisable
MovieClip.hitTestDisable

topmostLevel % º Ÿ f x U

topmostLevel:Boolean [read- write]

Scaleform : 2.1.50

true

false

"topmostLevel"

x Scaleform3.0.71
true

"topmostLevel"

topmost

x Scaleform3.0.72

;

topmost

A

B

topmost

A

B

"topmostLevel"

true

"topmostLevel"

SwapDepth

ActionScript

:
TextField.topmostLevel
MovieClip.topmostLevel

focusGroupMask %o Ÿ f x U

focusGroupMask : Number

Scaleform : 3.3.84

8 % -

setControllerFocusGroup

	button1	0	movieclip2
0	1		

Selection.setControllerFocusGroup(0, 0);
Selection.setControllerFocusGroup(1, 1);

button1.focusGroupMask = 0x1; // bit 0 –
movieclip2.focusGroupMask = 0x1 | 0x2 //

focus group 0
bits 0 and 1 –

focus groups 0 and 1

focusGroupMask

4 MovieClip a > k Z a k x ¥ i ™ ¥

hitTest () “ o u {

```
public hitTest(x:Number, y:Number, [shapeFlag:Boolean],
    [ignoreInvisibleChildren:Boolean]):Boolean
```

Scaleform : 2.1.51

3 hitTest 1 x, y

hitTest Flash

“true” (_visible false)

_alpha

Ã Ů Ó î ±

ignoreInvisibleChildren:Boolean – true

hitTestDisable % ¨ Ÿ f x U

hitTestDisable:Boolean [read- write]

Scaleform : 1.2.32

true MovieClip.hitTest

false

:

TextField.hitTestDisable

Button.hitTestDisable

noAdvance % ¨ Ÿ f x U

noAdvance : Boolean

Scaleform : 2.1.52

true

Flash

ActionScript

Scaleform Flash

“true”

:

_global.noInvisibleAdvance

topmostLevel %o ÿ f x U

topmostLevel:Boolean [read- write]

Scaleform : 2.1.50

 true

 false

"topmostLevel"

 x Scaleform3.0.71 "topmostLevel"

 true

topmost

;

 x Scaleform3.0.72 topmost

 A

 B

topmost

 A

 B

 "topmostLevel"

 true

 "topmostLevel"

SwapDepth

ActionScript

 :

 TextField.topmostLevel

 Button.topmostLevel

rendererString %o ÿ f x U

rendererString:String [read- write]

Scaleform : 2.2.55

MovieClip

ActionScript

[:

 myMovieInstance.rendererString = "SHADER_Blur"

rendererFloat %o f x U

rendererFloat:Number [read- write]

Scaleform : 2.2.55

MovieClip

ActionScript

```
[:  
    myMovieInstance.rendererFloat = 4;                    //    [FØG FÇ++F÷Fÿ 4 ò ÍF÷Fé
```

disableBatching %o f x U

disableBatching:Boolean

Scaleform : 4.0.17

focusGroupMask %o f x U

focusGroupMask : Number

Scaleform version: 3.3.84

8 % -

setControllerFocusGroup

	button1	0	movieclip2
0	1		

```
Selection.setControllerFocusGroup(0, 0);  
Selection.setControllerFocusGroup(1, 1);
```

```
button1.focusGroupMask = 0x1; // bit 0 –                    focus group 0  
movieclip2.focusGroupMask = 0x1 | 0x2 // bits 0 and 1 –                    focus groups 0 and 1
```

focusGroupMask

5 NetStream a k Z a k x ¥ i ™ ¥

Scaleform
NetStream

onMetaData V < ¥ z • ¥ { >

onMetaData = function(info:Object) { }

Scaleform : 3.0.63

Scaleform 2

onMetaData 2

1. audioTracks –

audioTracks :
 channelsNumber – (1- 2-
 6 – 5.1)
 totalSamples –
 trackIndex –
 NetStream.audioTrack (
)
 sampleRate –

2. subtitleTracksNumber –

audioTrack % ¨ f x U

audioTrack:Number [read- write]

Scaleform : 3.0.63

/

```
[:  
    var ns:NetStream = new NetStream(nc);  
    var audioTracks;  
  
    ns.onMetaData = function(info:Object)  
    {
```

```

audioTracks = info.audioTracks;
}

```

```

if (audioTracks != undefined && audioTracks.length > 0)
    ns.audioTrack = audioTracks[0].trackIndex;

```

subtitleTrack %o Ÿ f x U

subtitleTrack: Number [read- write]

Scaleform : 3.0.63

0

onSubtitle V < ¥ z • ¥ { >

onSubtitle = function(msg:String) {}

Scaleform : 3.0.63

[:

```

var ns:NetStream = new NetStream(nc);
var subtitlesNumber = 0;

ns.onMetaData = function(info:Object)
{
    subtitlesNumber = info.subtitleTracksNumber;
}

ns.onSubtitle = function(msg:String)
{
    sbTextField.text = msg;
}
if (subtitlesNumber > 0)
    ns.subtitleTrack = 1;

```

setNumberOfFramePools Ý /

public function setNumberOfFramePools(pools : Number) : Void

Scaleform : 3.0.68

“frame pools”

CPU

1

NetStream.play()call

setReloadThresholdTime Ý /

public function setReloadThresholdTime(reloadTime : Number) : Void

Scaleform : 3.0.68

Scaleform

0.8

NetStream

NetStream.setBufferTime()

2

1

1

/

NetStream.play()

6 Selection a > k Z a k x ¥ i ™ ¥

Selection

Scaleform Selection

ActionScript 1.0

Selection.captureFocus() Selection
 ActionScript 2.0
 ActionScript 2.0
 : Selection["captureFocus"]()

alwaysEnableArrowKeys - Ó %ö ÿ f x U

alwaysEnableArrowKeys : Boolean

Scaleform : 1.2.34

"_focusrect" false
 ()
 "_focusrect = false;" Flash
 alwaysEnableArrowKeys
 "true"

[:
 Selection["alwaysEnableArrowKeys"] = true;

alwaysEnableKeyPress - Ó %ö ÿ f x U

alwaysEnableKeyPress : Boolean

Scaleform : 3.0.63

"_focusrect" false
 Enter onPress / onRelease ()
 Flash "_focusrect = false;"
 alwaysEnableKeyPress "true"

[:
 Selection["alwaysEnableKeyPress"] = true;

captureFocus ()

public function captureFocus([doCapture:Boolean, controllerIdx:Number]) : void

Scaleform : 1.2.34

doCapture true () captureFocus
Tab

doCapture false captureFocus

Ã Û Ó î ±

doCapture:Boolean -

captureFocus

true

controllerIdx:Number -

0

[:

Selection["captureFocus"](); // "true" G" ûFéFþFø FèF÷Fé
Selection["captureFocus"](false);

disableFocusAutoRelease - Ó % ÿ f x U

disableFocusAutoRelease : Boolean

Scaleform : 1.2.34

(Flash

)

(

)
true

[:

Selection["disableFocusAutoRelease"] = true;

disableFocusKeys - Ó % ÿ f x U

disableFocusKeys : Boolean

Scaleform : 2.2.58

(Tab Shift-Tab)

```
[:  
    Selection["disableFocusKeys"] = true;  
    :  
    moveFocus()
```

disableFocusRolloverEvent ¯ Ó % ¨ ÿ f x U

disableFocusRolloverEvent : Boolean

Scaleform : 2.0.37

rollover/out

rollover/rollout true

```
[:  
    Selection["disableFocusRolloverEvent "] = true;
```

modalClip ¯ Ó % ¨ ÿ f x U

modalClip : MovieClip

Scaleform : 2.2.58

(Tab Shift-Tab
Tab)

```
[:  
    Selection["modalClip"] = _root.mc;  
    ...  
    Selection["modalClip"] = undefined;  
    :  
    disableFocusKeys  
    moveFocus()
```

moveFocus ()

```
public function moveFocus(keyToSimulate : String [, startFromMovie:Object,  
    includeFocusEnabledChars : Boolean = False, controllerIdx :  
    Number]) : Object
```

Scaleform : 2.2.58

(Tab Shift-Tab)

disableFocusKeys

modalClip

Ã Û Ó î ±

keyToSimulate:String – “up” “down” “left” “right” “tab” “shifttab”

startFromMovie: Object – moveFocus

null

3

includeFocusEnabledChars:Boolean – moveFocus focusEnabled
tabEnabled / tabIndex

false

tabEnabled / tabIndex

controllerIdx:Number –

0

[:

```
Selection["moveFocus"]("up");  
Selection["moveFocus"]("tab", _root.mc);  
Selection["moveFocus"]("tab", _root.mc, true);  
Selection["moveFocus"]("tab", null, true);
```

` ~ ¡

undefined

:
disableFocusKeys
modalClip

findFocus() k q x U u a “ o u {

```
public function findFocus(keyToSimulate : String [, parentMovie:Object, loop :
                        Boolean, startFromMovie:Object, includeFocusEnabledChars : Boolean,
                        controllerIndex : Number]) : Object
```

Scaleform : 3.3.84

TAB Shift TAB

disableFocusKeys

setModalClip/getModalClip

keyToSimulate:String – “right” “tab” “shifftab” “up” “down” “left”

parentMovie -

loop NULL -

Boolean

down findFocus null false

startFromMovie:Object – true
– findFocus

NULL

includeFocusEnabledChars:Boole an – focusEnabled
tabEnabled/tabIndex
moveFocus
false
tabEnabled/tabIndex

controllerIndex -

を組み合わせる

複数コントローラのフォーカスをサポート

var a = Selection["findFocus"]("up");

null

disableFocusKeys
 setModalClip
 getModalClip

setModalClip() k q x U u a “ o u {

public function setModalClip(modalClip : Object, controllerIndex : Number)

Scaleform : 3.3.84

TAB Shift TAB

modalClip	—
controllerIndex	-

getModalClip() k q x U u a “ o u {

public function getModalClip(controllerIndex : Number) : Object

Scaleform : 3.3.84

controllerIndex	-
-----------------	---

setControllerFocusGroup() k q x U u a “ o u {

public function setControllerFocusGroup(controllerIndex : Number, focusGroupIdx : Number) : Boolean

Scaleform : 3.3.84

controllerIndex	0
-----------------	---

2

	setControllerFocusGroup (1,1)	1
	setControllerFocusGroup (1,0)	
0	1	

controllerIndex	—
focusGroupIdx	—

Selection["setControllerFocusGroup"](0,0);
Selection["setControllerFocusGroup"](1,1);
Selection["setControllerFocusGroup"](2,1);

true

getControllerFocusGroup() k q x U u a “ o u {

public function getControllerFocusGroup (controllerIndex : Number) : Number

Scaleform : 3.3.84

controllerIndex	—
-----------------	---

getFocusArray() k q x U u a “ o u {

public function getFocusArray(mc : Object) : Array

Scaleform : 3.3.84

mc	-
----	---

getFocusBitmask() k q x U u a “ o u {

public function getFocusBitmask(mc : Object) : Number

Scaleform : 3.3.84

mc -

getControllerMaskByFocusGroup() k q x U u a “ o u {

public function getControllerMaskByFocusGroup (focusGroupIdx : Number) : Number

Scaleform : 3.3.84

focusGroupIdx -

numFocusGroups %o Ÿ f x U

numFocusGroups : Number

Scaleform : 3.3.84

setControllerFocusGroup 0 3 numFocusGroups 2

7 TextField a > k Z a k x ¥ i ™ ¥

Scaleform
 TextField
 Flash 9 TextField
 IME
)

Flash 8
 (

7.1 3 1 Ó) š

autoFit % Ÿ f x U

autoFit:Boolean [read- write]

Scaleform : 2.0.39

/

false

appendText () “ o u {

public function appendText(newText:String):void

Scaleform : 2.0.37

newText

(+=) (my_txt.text +=

appendText)

Ã Ů Ó î ±

newText:String –

:

appendHtml()


```

newHtml      HTML
      htmlText      (+ =      (txt.htmlText += moreHtml      )
      htmlText      +=      HTML
HTML      HTML      HTML      HTML
      newHtml      HTML      (newHtml      HTML
      +=      )

```

```
newHtml:String - HTML
:
appendText()
```

caretIndex %o Y f x U

caretIndex: Number [read-only]

Scaleform : 2.0.37

```

Selection.getCaretIndex()
0
caretIndex

```

```
Selection.getCaretIndex() -1
( 0 )
:
```

getCharBoundaries () “ o u {

public function getCharBoundaries(charIndex: Number): flash.geom.Rectangle

Scaleform : 2.0.37

‘a’ ‘w’ ‘k’): (

Baywio/ks

Ã Ů Ó î ±

charIndex:Number – (0
1)

` ~ ¡

flash.geom.Rectangle – x y
(0,0)

:
getExactCharBoundaries()

getExactCharBoundaries () “ o u {

public function getExactCharBoundaries(charIndex:Number): flash.geom.Rectangle

Scaleform : 2.0.37

(getCharBoundaries() ‘a’ ‘w’ ‘k’)

:

Baywio/ks

Ã Ů Ó î ±

charIndex:Number – (0
1)

~Í

flash.geom.Rectangle – x y
(0,0)

:
getCharBoundaries()

getCharIndexAtPoint () “ o u {

public function getCharIndexAtPoint(x:Number, y:Number): Number

Scaleform : 2.0.37

x y

Ã Ô Ó î ±

x:Number – Q b e Ð b x Û †
y:Number – Q b e Ð b y Û †

~Í

Number – (0 1
-1

getFirstCharInParagraph () “ o u {

public function getFirstCharInParagraph(charIndex:Number): Number

Scaleform : 2.0.37

charIndex

Ã Ô Ó î ±

charIndex:Number – (0
1)

~Í

Number –

getLineIndexAtPoint () “ o u {

public function getLineIndexAtPoint(x:Number, y:Number): Number

Scaleform : 2.0.37

x y

Ã Ũ Ó î ±

x:Number – Q b/œ b x Û †

y:Number – Q b/œ b y Û †

` ~ ¡

Number – (0 1) -1

getLineLength() “ o u {

public function getLineLength(lineIndex:Number): Number

Scaleform : 2.0.37

Ã Ũ Ó î ±

lineIndex:Number – (0) 1

` ~ ¡

Number –

getLineMetrics() “ o u {

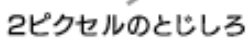
public function getLineMetrics(lineIndex:Number): Object

Scaleform : 2.0.43

:

ascent : Number

descent : Number

$$\left(\begin{array}{c} \text{ } \end{array} \right)$$
$$\left(\begin{array}{c} \text{ } \end{array} \right)$$


(0

getLineOffset () “ o u {

public function getLineOffset(lineIndex:Number): Number

Scaleform : 2.0.37

lineIndex

Ã Ů Ó î ±

lineIndex:Number – (0
1)

` ~ Í

Number –

getLineText () “ o u {

public function getLineText(lineIndex:Number): String

Scaleform : 2.0.43

lineIndex

Ã Ů Ó î ±

lineIndex:Number – (0
1)

` ~ Í

String –

hitTestDisable % Ÿ f x U

hitTestDisable:Boolean [read- write]

Scaleform : 1.2.32

true MovieClip.hitTest

false

noTranslate %o Ÿ f x U

noTranslate:Boolean [read- write]

Scaleform : 1.2.34

true GfxTranslator::Translate

false

numLines %o Ÿ f x U

numLines:Number [read- only]

Scaleform : 2.0.43

wordWrap

true

topmostLevel %o Ÿ f x U

topmostLevel:Boolean [read- write]

Scaleform : 2.1.50

true

false

"topmostLevel"

x Scaleform3.0.71
 true

"topmostLevel"

topmost

x Scaleform3.0.72
 ;

topmost

A

B

topmost

A

B

"topmostLevel"

true

"topmostLevel"

SwapDepth

ActionScript

:


```

:
selectionBkgColor
selectionTextColor
inactiveSelectionBkgColor
inactiveSelectionTextColor

```

```
copyToClipboard () " o u {
```

```
public function copyToClipboard([richClipboard:Boolean], [startIndex:Number],
[endIndex:Number]):void
```

Scaleform : 2.1.45

Ã Ů Ó î ±

```
richClipboard:Boolean - true
useRichTextClipboard
```

```
startIndex:Number -
selectionBeginIndex
```

```
endIndex:Number -
selectionEndIndex
```

```

:
useRichTextClipboard
selectionBeginIndex
selectionEndIndex
cutToClipboard()
pasteFromClipboard()

```

```
cutToClipboard () " o u {
```

```
public function cutToClipboard([richClipboard:Boolean], [startIndex:Number],
[endIndex:Number]):void
```

Scaleform : 2.1.45

Ã Ů Ó î ±

```
richClipboard: Boolean - true
useRichTextClipboard
```

startIndex: Number –

selectionBeginIndex

endIndex: Number –

selectionEndIndex

:

useRichTextClipboard

selectionBeginIndex

selectionEndIndex

copyToClipboard()

pasteFromClipboard()

inactiveSelectionBkgColor %Y f x U

inactiveSelectionBkgColor: Number [read- write]

Scaleform : 2.1.45

alwaysShowSelection

true

0xAARRGGBB

AA

16

[0...255] RR

16

[0...255] BB

16

[0...255]

GG

16

[0...255]

“0”

“0”

Flash

0xFFFF0000 (0xFF(255))

“backgroundColor ”

0xFF0000

F1

6õ4 8o% :

alwaysShowSelection

inactiveSelectionTextColor

selectionBkgColor

selectionTextColor

inactiveSelectionTextColor %Y f x U

inactiveSelectionTextColor: Number [read- write]

Scaleform : 2.1.45

alwaysShowSelection

true

0xAARRGGBB

AA

16

[0..255] RR

```

GG          16      16      [0..255] BB          16      [0..255]
          [0..255]
          "0"
          "0"
Flash
          0xFFFF0000 (          0xFF(255)
)          "textColor"          0xFF0000

```

```

:
alwaysShowSelection
inactiveSelectionBkgColor
selectionBkgColor
selectionTextColor

```

```
noAutoSelection    %o Ÿ f x U
```

```
noAutoSelection:Boolean [read-      write]
```

```
Scaleform          : 2.1.45
```

```
true
```

```
false
```

```
pasteFromClipboard ()    " o u {
```

```
public function pasteFromClipboard([richClipboard:Boolean], [startIndex:Number],
[endIndex:Number]):void
```

```
Scaleform          : 2.1.45
```

```
Ã Ů Ó î ±
```

```
richClipboard:Boolean    - true
```

```
useRichTextClipboard
```

```
startIndex:Number    -
```

```
selectionBeginIndex
```

```
endIndex:Number    -
```

```
selectionEndIndex
```

```

:
useRichTextClipboard
selectionBeginIndex
selectionEndIndex
copyToClipboard()
cutToClipboard()

```

selectionBeginIndex %o Ÿ f x U

selectionBeginIndex: Number [read-only]

Scaleform : 2.1.45

caretIndex

```

Selection.getBeginIndex()
selectionBeginIndex
Selection.getBeginIndex()

```

-1

```

:
caretIndex
selectionEndIndex
Selection.getBeginIndex()

```

selectionEndIndex %o Ÿ f x U

selectionEndIndex: Number [read-only]

Scaleform : 2.1.45

caretIndex

```

Selection.getEndIndex()
selectionEndIndex
Selection.getEndIndex()

```

-1

```

:
caretIndex
selectionBeginIndex
Selection.getEndIndex()

```


useRichTextClipboard %_ f x U

useRichTextClipboard:Boolean [read- write]

Scaleform : 2.1.45

Scaleform & true
(
) useRichTextClipboard true
false

7.3 x_k z g V l' ·\$k

() Scaleform

fontScaleFactor %_ f x U

fontScaleFactor:Number [read- write]

Scaleform : 2.0.43

fontScaleFactor

1.0

textAutoSize %_ f x U

textAutoSize:String [read- write]

Scaleform : 2.0.43

textAutoSize
"none" "shrink" "fit" ("shrink" "fit")

(5 pt)

textAutoSize "fit"

"shrink"

"none"

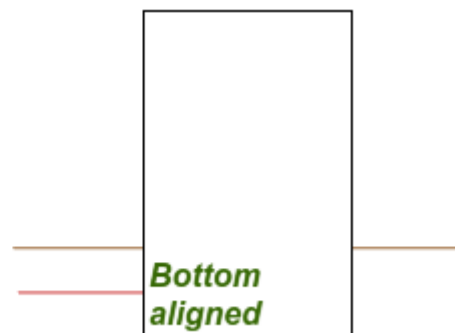
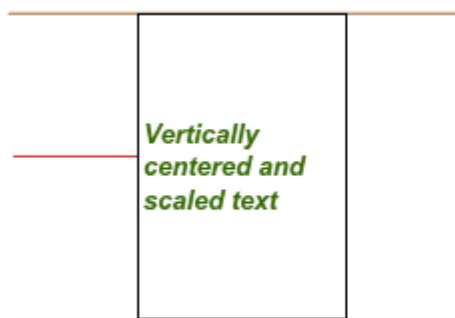


verticalAlign %o Ÿ f x U

verticalAlign:String [read- write]

Scaleform : 2.0.43

"none" ("top" "none") "bottom" "center" "bottom" "center"
()



"none"

:
verticalAutoSize
TextField.align

```
verticalAutoSize    %o Ÿ f x U
verticalAutoSize:String [read-      write]
```

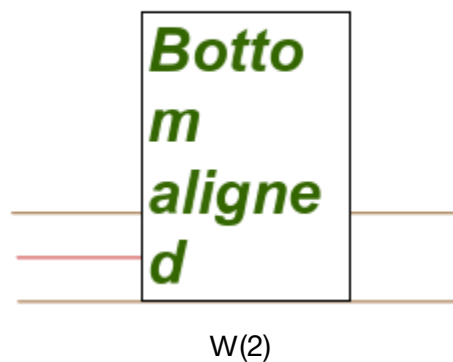
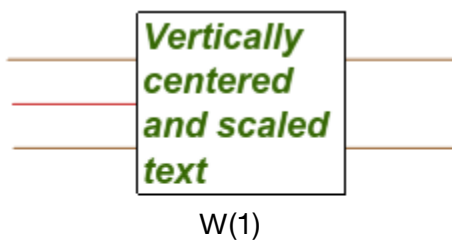
Scaleform : 2.0.43

```
    ) "top" "bottom" "center"    verticalAutoSize "none" (
                                   "none" (
                                   )
```

```
verticalAutoSize    "top"                wordWrap    true
                    (
                    )    TextField.autoSize
```

```
verticalAutoSize    "center "
                    (
                    )    1
```

```
verticalAutoSize    "bottom "
                    (
                    2    )
```



"none "

```
:
verticalAlign
TextField.autoSize
```


7.4 HTML Z a k x ¥ i ™ ¥

```

Scaleform HTML
<FONT> q b ALPHA 0
ALPHA="#xx"
00 FF (0 255) 0
ALPHA 255 HTML COLOR
<IMG> q b
Scaleform <IMG>
Scaleform
Web
("http://" "file://" ) 5.6
"myImage" <IMG> HTML :
t.htmlText = "<p align='right'>abra<img src='myImage' width='20' height='30'
align='baseline' vspace='- 10'>bed232</p>";
IMG :
src -
JPEG GIF PNG SWF
ID img://
Scaleform::ImageCreator:: LoadImage
width -
height -
align -
"baseline "
vspace - "baseline "
:
```

id – (Flash Player SWF)

align –

hspace -

SWF IMG
Flash
Flash SWF
IMG

Flash SWF

7.5 $i \cdot \{X \cdot \ddot{Y} - j$

()

blurX ' blurY %o $\ddot{Y} f \times U$

blurX:Number [read- write]
blurY:Number [read- write]

Scaleform : 2.0.41

() 0 15.9 ()

0

:
blurStrength

blurStrength %o $\ddot{Y} f \times U$

blurStrength:Number [read- write]

Scaleform : 2.0.41

0 15.9 ()

0

:
blurX
blurY

shadowAlpha % ƒ x U

shadowAlpha:Number [read- write]

Scaleform : 2.0.41

0.00 1.00 25

25%

:
shadowColor

shadowAngle % ƒ x U

shadowAngle:Number [read- write]

Scaleform : 2.0.41

) DropShadowFilter 0 360° (45

:
DropShadowFilter

shadowBlurX % ƒ x U shadowBlurY % ƒ x U

:Number [read- write]

shadowBlurY:Number [read- write]

Scaleform : 2.0.41

() 0 15.9 ()

0

:

shadowStrength

shadowColor %Y f x U

shadowColor: Number [read- write]

Scaleform : 1.1.31

```

0xRRGGBB      ( shadowStyle )
[0..255]  BB      RR      16
      [0..255]  GG      16

:
shadowStyle

```

shadowDistance %o $\ddot{Y}$ $f \times U$

shadowDistance: Number [read- write]

Scaleform : 2.0.41

DropShadowFilter

DropShadowFilter

shadowHideObject %o $\ddot{Y}$ f x U

shadowHideObject: Boolean [read- write]

Scaleform : 2.0.41

true
 false ()

shadowKnockOut %o $\ddot{Y}$ f x U

shadowKnockOut:Boolean [read- write]

Scaleform : 2.0.41

(true) false ()

shadowQuality %o Ÿ f x U
shadowQuality:Number [read- write]

Scaleform : 2.0.41

Flash

1 2

1 2

shadowStrength %o Ÿ f x U

shadowStrength:Number [read- write]

Scaleform : 2.0.41

0 15.9 ()

0

:
shadowBlurX
shadowBlurY

shadowStyle %o Ÿ f x U

shadowStyle:String [read- write]

Scaleform : 1.1.31

shadowColor
shadowStyle

:

field.shadowStyle = "s{ 0,- 1.5}{- 1.4,1.2}{1.4,1.2}t{ 0,0}";
field.shadowColor = 0xff0000;

 "s" shadowColor
 "t" TextField.textColor
 "s" "t"

1 "t"
 "t{0,0}"

```

:
shadowColor
TextField.textColor

```

7.6 V “ 4 j - ¼

```
setImageSubstitutions () “ o u {
```

```

public setImageSubstitutions(substInfoArr:Array) : void
public setImageSubstitutions(substInfo:Object) : void
public setImageSubstitutions(null) : void

```

Scaleform : 2.0.38

SWF

SWF

:

- 1.
2. (Windows) control-
[]
3. [ActionScript] []
(myImage) []
4. [OK]

BitmapData

ActionScript :

```

import flash.display.BitmapData;
var imageBmp:BitmapData = BitmapData.loadBitmap("myImage");

```

```

: (import flash.display.BitmapData;
flash.display.BitmapData )
"undefined " !

```

ID

Object

:

subString:String

15

image : BitmapData

width : Number

()

height : Number

()

baseLineY : Number

Y

(

)

id : String

"updateImageSubstitution"

ID

"substInfoArr"

"substInfo"

setImageSubstitutions(substInfo:Object)

setImageSubstitutions(substInfoArr:Array)

setImageSubstitutions

setImageSubstitutions(null)

setImageSubstitutions

ActionScript

ActionScript

Ã Ů Ó î ±

substInfoArr:Array -

substInfo:Object -

null -

()

()

```

:
updateImageSubstitution()

```

```

[:

```

```

var b1 = BitmapData.loadBitmap("smile1");
var b2 = BitmapData.loadBitmap("smile2");
var b3 = BitmapData.loadBitmap("smile3");
var a = new Array;
a[0] = { subString:"=", image:b1, baseLineY:35, width:20, height:20, id:"sm=" } };
a[1] = { subString:"-      )", image:b2, baseLineY:20, id:"sm:-      )" };
a[2] = { subString:":\      \ ", image:b3, baseLineY:35, height:100 };
a[3] = { subString:"-      \ \ ", image:b1 };
t.setImageSubstitutions(a);

```

```

"=")

```

```

"smile1"

```

```

updateImageSubstitution () " o u {

```

```

public updateImageSubstitution(id:String, image:BitmapData) : void

```

```

Scaleform : 2.0.38

```

```

setImageSubstitutions

```

```

Ã Ů Ó î ±

```

```

id:String -

```

```

ID setImageSubstitutions

```

```

image:BitmapData - id
null

```

```

: setImageSubstitutions()

```

```

[:

```

```

t.updateImageSubstitution("sm=", b3);

```

```

onEnterFrame

```

```

setInterval updateImageSubstitution

```

```

var phase = 0;
var b1a = BitmapData.loadBitmap("smile1a");

```



```

var b2a = BitmapData.loadBitmap("smile2a");

onEnterFrame = function()
{
    if (phase % 10 == 0)
    {
        if (phase % 20 == 0)
        {
            _root.t.updateImageSubstitution("sm=", b1);
            _root.t.updateImageSubstitution("sm:-", b2);
        }
        else
        {
            _root.t.updateImageSubstitution("sm=", b1a);
            _root.t.updateImageSubstitution("sm:-", b2a);
        }
    }
    ++phase;
}

```

7.7 IME - g • 4 z

IME

disableIME %o Ÿ f x U

disableIME:Boolean [read- write]

Scaleform b Â î ª Ò å: 2.1.50 è7F

true IME

false

```
getIMECompositionStringStyle () " o u {
public function getIMECompositionStringStyle(categoryName:String): Object
```

Scaleform : 2.1.50

```
IME
categoryName setIMECompositionStringStyle
```

```

    categoryName:String - IME (setIMECompositionStringStyle
    )
```

```

    Object - (setIMECompositionStringStyle
    )
```

```

    :
    setIMECompositionStringStyle
    disableIME
```

```
setIMECompositionStringStyle () " o u {
public function setIMECompositionStringStyle(categoryName:String,
styleDescriptor:Object): Number
```

Scaleform : 2.1.50

```
IME
categoryName
```

```

    " compositionSegment" -
    " clauseSegment" -
    " convertedSegment" -
    " phraseLengthAdj" -
    " lowConfSegment" -
```

```
styleDescriptor Object :
```

```
textColor : Number
```

```
backgroundColor : Number
```

• •

disableIME

8 TextFormat a > k Z a k x ¥ i ™ ¥

TextFormat

alpha % º Ÿ f x U

alpha:Number [read- write]

Scaleform : 2.0.41

0 100 (%)

Flash TextFormat.color

:

TextFormat

HTML

9 Stage a > k Z a k x ¥ i ™ ¥

Stage

Scaleform

“visibleRect” “safeRect” “originalRect”
“onResize”

ActionScript 1.0 Stage.visibleRect

ActionScript 2.0

2.0

Stage[“visibleRect”].

ActionScript

:

visibleRect % º ¨ f x U

visibleRect:flash.geom.Rectangle [read-only]

Scaleform : 2.2.56

topLeft

:

safeRect

originalRect

safeRect % º ¨ f x U

safeRect:flash.geom.Rectangle [read-only]

Scaleform : 2.2.56

GFX::Movie: :SetSafeRect

“visibleRect”

:

visibleRect

originalRect

10 Array a > k Z a k x ¥ i ™ ¥

Ÿ c 4 ••¥ - ' 8 y

Flash	Array.sort	Array.sortOn	Unicode
	Unicode		(
	)	'z'	['á',
'z', 'a']	Flash		['a', 'z',
'á']			['a', 'á', 'z']
	Scaleform	Array.sort/sortOn	1 Array.LOCALE
	:		

```
var arr = ['á', 'z', 'a'];
var newArr1 = arr.sort(); // returns ['a', 'z', 'á']
var newArrLoc = arr.sort(Array.LOCALE); // returns ['a', 'á', 'z']
var newArrRev = arr.sort(Array.LOCALE | Array.DESENDING); // ['z', 'á', 'a']
```

:

```
var arr = ['Á', 'z', 'a'];
var a = arr.sort(Array.LOCALE | Array.CASEINSENSITIVE); // returns ['a', 'Á', 'z']
```

Array.LOCALE

Unicode

Scaleform : 3.0.65

String.localeCompare

11 String a > k Z a k x ¥ i ™ ¥

localeCompare () “ o u {

2

0

Array

Unicode

12 Video a > k Z a k x ¥ i ™ ¥

Flash

Video

NetStream

Scaleform

Video

NetStream

clipRect % ¨ f x U

clipRect: flash.geom.Rectangle [read- write]

Scaleform : 3.0.70

[:

```
video.clipRect = new flash.geom.Rectangle(10, 20, 100, 100);
```

13 3Di a > k Z a k x ¥ i ™ ¥

3Di Scaleform AS2

3Di ActionScript

Scaleform : 3.2.82

_z

Z

0

_zscale

Z

100

_xrotation

X

0

_yrotation

Y

0

_matrix3d

16

NULL

4 × 4

3D

3D
2D

_perspfov

Field Of View

55°

FOV

0

180

ActionScript

gfxExtensions true

14 b Ÿ 4 , • Z a k x ¥ i ™ ¥

noInvisibleAdvance %o Ÿ f x U

noInvisibleAdvance : Boolean

Scaleform : 2.1.52

true

Flash SWF ("true"
ActionScript)
Scaleform Flash

:
MovieClip.noAdvance

imecommand Ÿ /

imecommand(command:String, parameters:String) : Void

Scaleform : 2.1.50

fscommand Scaleform IME
Scaleform

getIMECandidateListStyle () Ÿ /

public function getIMECompositionStringStyle(categoryName:String): Object

Scaleform : 2.1.50

IME

setIMECandidateListStyle

` ~ Í

Object – Ã/õ Ü « ° b • Ů î0¿ 0°3Ù Ê

6õ4 8o% :

setIMECandidateListStyle
TextField.disableIME

setIMECandidateListStyle () Ý /

public function setIMECandidateListStyle(styleDescriptor:Object)

Scaleform : 2.1.50

styleDescriptor Object :

styleObject.textColor	= 0x5EADFF;
styleObject.selectedTextColor	= 0xFFFFFFFF;
styleObject.fontSize	= 20;
styleObject.backgroundColor	= 0x001430;
styleObject.selectedTextBackgroundColor	= 0x2C5A99;
styleObject.indexBackgroundColor	= 0x12325A;
styleObject.selectedIndexBackgroundColor	= 0x2C5A99;
styleObject.readingWindowTextColor	= 0xFFFFFFFF;
styleObject.readingWindowBackgroundColor	= 0x001430;
styleObject.readingWindowFontSize	= 20;

textColor : Number

selectedTextColor : Number

fontSize : Number

backgroundColor : Number

seletedTextBackgroundColor : Number

indexBackgroundColor : Number

selectedIndexBackgroundColor : Number

readingWindowTextColor : Number

readingWindowBackgroundColor : Number

readingWindowFontSize : Number

	0xRRGGBB		RR
16	[0..255] BB	16	[0..255] GG
16	[0..255]		

:
getIMECandidateListStyle
TextField.disableIME

15 k q ¥ r 4 { “ o u { ' V < ¥ z • ¥ { > 4 Z a k x ¥ i ™ ¥

Key a > k

Key.getCode(controllerIdx:Number)
Key.getAscii(controllerIdx:Number)
Key.isDown(controllerIdx:Number)
Key.isToggled(controllerIdx:Number)

Key

controllerIdx 0

Key.onKeyDown(controllerIdx:Number)

onKeyDown

Key

Selection a > k

Selection.setFocus(ch:Object, controllerIdx:Number)
Selection.getFocus(controllerIdx:Number)
Selection.getBeginIndex(controllerIdx:Number)
Selection.getEndIndex(controllerIdx:Number)
Selection.setSelection(start:Number, end:Number, controllerIdx:Number)
Selection.getCaretIndex(controllerIdx:Number)

Selection

controllerIdx 0

Selection.onSetFocus(old:Object, new:Object, controllerIdx:Number)

onSetFocus

Selection

MovieClip/Button/TextField

MovieClip.onSetFocus(old:Object, controllerIdx:Number)
MovieClip.onKillFocus(new:Object, controllerIdx:Number)
Button.onSetFocus(old:Object, controllerIdx:Number)
Button.onKillFocus(new:Object, controllerIdx:Number)
TextField.onSetFocus(old:Object, controllerIdx:Number)
TextField.onKillFocus(new:Object, controllerIdx:Number)
TextField.onChange(controllerIdx:Number)

MovieClip/Button/TextField

System.capabilities

System.capabilities.numControllers —